

Inflation Dynamics - Computational Notebook

Haoyang Sun

Abstract

This book presents computation details of estimating inflation dynamics.

This notebook presents computation details of estimating inflation dynamics.

So far we have covered estimation of

1. GARCH
2. EGARCH
3. GJR-GARCH
4. SPARCH
5. Regime-switching GARCH
6. BEGE GARCH

State variable definitions:

1. π_t : inflation.
2. $\mathcal{F}_{t-1}$: filtration at time t.
3. SPF_t : professional forecast observed in $\mathcal{F}_{t-1}$.
4. $\hat{\pi}_t$: expected inflation $E[\pi_t \mid \mathcal{F}_{t-1}]$.
5. u_t : inflation innovation (residual), defined as $u_t := \pi_t - \hat{\pi}_t$.
6. u_t^+ : positive residual, defined as $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$
7. u_t^- : negative residual, defined as $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$

1. INFLATION SUMMARY STATISTICS

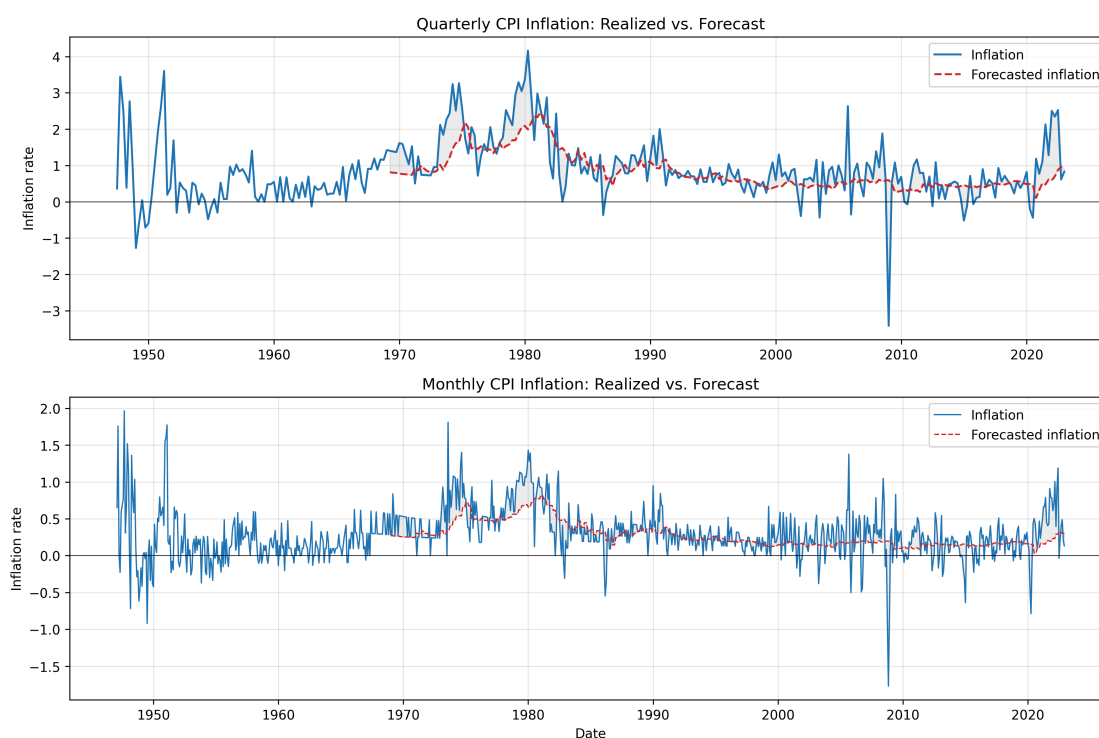


Figure 1: Realized inflation and profesional forecast over quarter and month frequency.

1.a. Quarterly Sheet

	Inflation	Forecasted inflation	Inflation shock
Date Start	1947Q2	1969Q1	1969Q1
Date End	2022Q4	2022Q4	2022Q4
#Observations	303	216	216
Mean	0.8686	0.8319	0.1618
Median	0.7246	0.6310	0.1112
Std	0.8640	0.4957	0.6633
Min	-3.4170	0.1048	-4.0117
Max	4.1612	2.4566	2.1729
P5	-0.1861	0.3482	-0.7677
P25	0.4187	0.4748	-0.1666
P75	1.1758	1.0147	0.4738
P95	2.5669	2.0006	1.2571
Skewness	0.6203	1.3670	-0.7204
Kurtosis	3.0886	1.2257	7.3167
AC(1)	0.5714	0.9717	0.2778
AC(2)	0.4923	0.9473	0.1338

	Inflation	Forecasted inflation	Inflation shock
AC(4)	0.3780	0.8929	0.0910
AC(12)	0.2842	0.6949	-0.0003

1.b. Monthly Sheet

	Inflation	Forecasted inflation	Inflation shock
Date Start	1947-02	1969-01	1969-01
Date End	2022-12	2022-12	2022-12
#Observations	911	648	648
Mean	0.2901	0.2763	0.0532
Median	0.2463	0.2099	0.0414
Std	0.3452	0.1637	0.2770
Min	-1.7705	0.0349	-1.9684
Max	1.9643	0.8123	1.4582
P5	-0.1671	0.1159	-0.3443
P25	0.1006	0.1580	-0.0816
P75	0.4438	0.3371	0.1977
P95	0.9432	0.6689	0.4963
Skewness	0.5757	1.3541	-0.4679
Kurtosis	3.9681	1.1670	6.6287
AC(1)	0.5776	0.9906	0.4690
AC(2)	0.4159	0.9811	0.2108
AC(4)	0.3663	0.9636	0.1182
AC(12)	0.1853	0.8932	0.0007

1.c. The Number of Observations

To compare different models using log likelihood, we need to control the number of observations the same for all models. One thing that needs attention is that for same data set, the model with one lag has one more observation than the model with two lags.

Freedom-adjusted measure, e.g AIC and BIC, does not have this concern.

It is crucial to unify how to treat the lag variable, which could lead to different sample size and thus lead the likelihood function un-comparable.

In the 2023 version, I did a mistake removing two more observations in garch1 when cleaning the data. This is remaining mystery for haoyang that how many observations should include.

2. MEAN PROCESS SPECIFICATION

Four types of mean processes are of interest:

- **Constant:** $\pi_{t+1} = SPF_t + \mu_{t+1}$
- **ARX(1,1):** $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,1):** $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,2)¹:** $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

3. OLS RESULTS

I report OLS estimation results of three mean models with HAC standard error²:

ARX(1,1):

$$\hat{\pi}_{t+1} = 0.0720 + 0.2881 \pi_t + 0.7508 SPF_t \quad (1)$$

(0.087) (0.114) (0.168)

ARX(2,1):

$$\hat{\pi}_{t+1} = 0.0792 + 0.2793 \pi_t + 0.0728 \pi_{t-1} + 0.6661 SPF_t \quad (2)$$

(0.087) (0.102) (0.127) (0.257)

ARX(2,2):

$$\hat{\pi}_{t+1} = 0.0761 + 0.2880 \pi_t + 0.0799 \pi_{t-1} + 0.4720 SPF_t + 0.1793 SPF_{t-1} \quad (3)$$

(0.088) (0.109) (0.127) (0.440) (0.295)

¹In the GARCH family estimation exercise, we found that ARX(2,2) is always dominated by other three specifications. Thus, we stop estimating ARX(2,2) in BEGE estimation to saving computational resources.

²Standard errors are heteroskedasticity and autocorrelation robust (HAC) using 12 lags and without small sample correction.

4. GARCH

4.a. Model Specification

GARCH(p,q) model specification:

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i u_{t-i}^2 + \sum_{k=1}^q \beta_k \sigma_{t-k}^2 \quad (4)$$

5. GARCH ESTIMATION IN PYTHON'S arch PACKAGE: SAMPLE USE AND BACKCASTING

5.a. How arch Uses the Sample

For a GARCH(p, q) model with N observations and residuals ε_t :

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2 \quad (5)$$

arch uses **all N observations** in the log-likelihood — it does not drop the first $\max(p, q)$. The unobserved pre-sample values are filled in via **backcasting**: a single value $\hat{\sigma}_{bc}^2$ initializes all required pre-sample quantities,

$$\hat{\varepsilon}_0^2 = \hat{\varepsilon}_{-1}^2 = \dots = \hat{\sigma}_0^2 = \hat{\sigma}_{-1}^2 = \dots = \hat{\sigma}_{bc}^2 \quad (6)$$

5.b. The Four Standard Backcasting Methods

1. Sample (unconditional) variance. Equal weighting of squared residuals:

$$\hat{\sigma}_{bc}^2 = \frac{1}{N} \sum_{t=1}^N \varepsilon_t^2 \quad (7)$$

Default in EViews and MATLAB.

2. Exponentially weighted (RiskMetrics-style) — arch default.

Geometrically decaying weights:

$$\hat{\sigma}_{bc}^2 = \frac{\sum_{t=1}^N \lambda^{t-1} \varepsilon_t^2}{\sum_{t=1}^N \lambda^{t-1}} \quad (8)$$

with $\lambda = 0.94$ in arch. Default in arch and rugarch (R).

3. Model-implied long-run variance (variance targeting). Unconditional variance from current parameters:

$$\hat{\sigma}_{bc}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j} \quad (9)$$

Undefined when $\sum \alpha_i + \sum \beta_j \geq 1$.

4. Fixed / user-specified value.

$$\hat{\sigma}_{bc}^2 = c \quad (10)$$

A constant supplied before optimization.

5.c. Why the Choice Rarely Matters

The influence of $\hat{\sigma}_{\text{bc}}^2$ on σ_t^2 decays geometrically as β^t . For typical $\beta \approx 0.85$, it falls below 1% by $t \approx 30$. Bollerslev (1986) showed the choice of starting values does not affect the asymptotic distribution of the MLE, and in practice the four methods yield estimates that differ by less than the standard errors.

5.d. Estimation Summary

5.e. Best Estimation Illustrations

6. GJR-GARCH SPECIFICATION

GJR GARCH model specification:

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i |u_{t-i}|^2 + \sum_{j=1}^o \gamma_j |u_{t-j}|^2 I[u_{t-j} < 0] + \sum_{k=1}^q \beta_k \sigma_{t-k}^2 \quad (11)$$

7. INFLATION DYNAMICS

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma, \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

Table 1: Parameter Bounds for Mean Process Specifications

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	—	—	—	—	—
ARX(1,1)	$(\min \pi_t, \max \pi_t)$	$(-0.999, 0.999)$	—	$(-10, 10)$	—
ARX(2,1)	$(\min \pi_t, \max \pi_t)$	$(-1.999, 1.999)$	$(-0.999, 0.999)$	$(-10, 10)$	—
ARX(2,2)	$(\min \pi_t, \max \pi_t)$	$(-1.999, 1.999)$	$(-0.999, 0.999)$	$(-10, 10)$	$(-10, 10)$

7.a. II. GARCH Model Summarization

Table 2: Model selection by AIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	359.4710	(1,1)	(1,1)	368.2429
t	(1,1)	(2,1)	343.1431	(2,1)	(1,1)	344.6254
Skew t	(1,1)	(2,1)	345.1312	(2,1)	(1,1)	346.5791
GED	(1,1)	(2,1)	345.1925	(1,1)	(1,1)	347.6934
Mix of Normal	(1,1)	(2,1)	345.8222	(2,1)	(2,1)	343.5023

Table 3: Model selection by BIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	381.4280	FC	(1,1)	383.4754
t	FC	(1,1)	367.8350	FC	(1,1)	367.1732
Skew t	FC	(1,1)	370.4242	FC	(1,1)	369.6649
GED	FC	(1,1)	371.8489	FC	(1,1)	370.9644
Mix of Normal	FC	(1,1)	373.7106	FC	(1,1)	371.0689

EGARCH model specification:

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \left(|e_{t-i}| - \sqrt{2/\pi} \right) + \sum_{j=1}^o \gamma_j e_{t-j} + \sum_{k=1}^q \beta_k \ln \sigma_{t-k}^2 \quad (12)$$

where $e_t = \varepsilon_t / \sigma_t$

8. BEGE DENSITY

BEGE density function $f_{BEGE}(\mu | p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation μ given the parameters $\{p, n, \sigma_p, \sigma_n\}$.

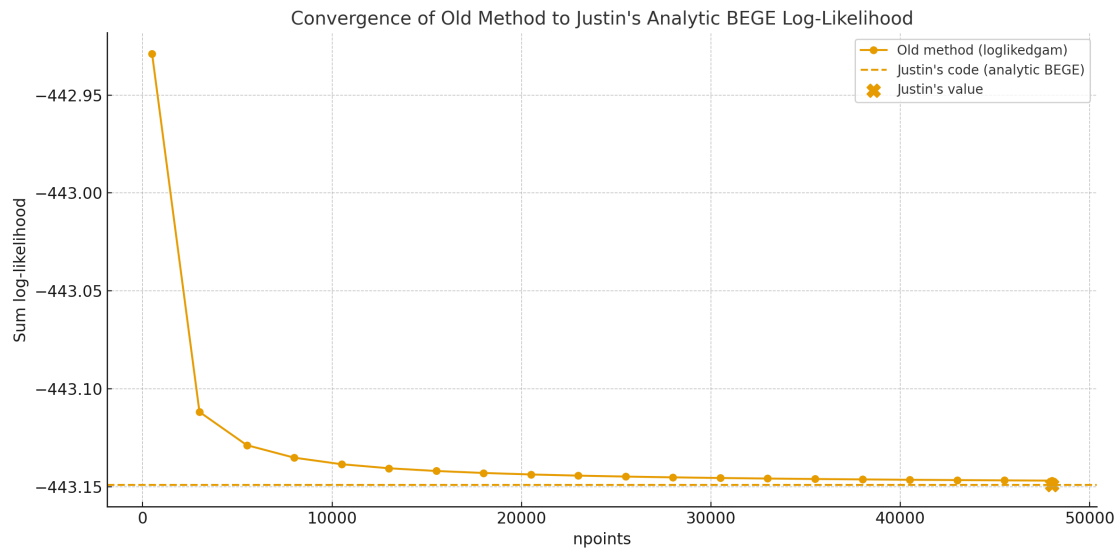
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \quad n = 7, \quad \sigma_p = 0.8, \quad \sigma_n = 1.2. \quad (13)$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function*.

The old numerical method was evaluated over a wide range of grid resolutions (up to 50,000 points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

9. BEGE GARCH FAMILY

9.a. Constant BEGE

The first model that I estimated is BEGE with invariant shape parameters, which is

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n\end{aligned}\tag{14}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1).\tag{15}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

9.b. Symmetric BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n,\end{aligned}\tag{16}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1).\tag{17}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned}p_t &= p_0 + \rho p_{t-1} + \frac{\phi^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho n_{t-1} + \frac{\phi^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_n^2} (u_{t-1}^-)^2\end{aligned}\tag{18}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho < 0.999$,
- $10^{-5} < \phi^+, \phi^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

9.c. Bad and Good Symmetric GARCH Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n)$:

$$\begin{aligned}p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p}{2\sigma_p^2} (u_{t-1})^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n}{2\sigma_n^2} (u_{t-1})^2\end{aligned}\tag{19}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$.

- $\max\{p_t, n_t\} < 200$

9.d. Inflation and Deflation Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \quad (20)$$

According to the emails, I have set the constraints below.

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

9.e. Full BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n, \end{aligned} \quad (21)$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \quad (22)$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi_p^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \quad (23)$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho_p, \rho_n < 0.999$,
- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$ where $\text{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.
- $\max\{p_t, n_t\} < 200$.

10. RESULT SUMMARY

Overall, most of the estimations with random initial values converge. I randomly sampled 100,000 starting values for the constant BEGE model, and sampled 10,000 for symmetric BEGE because it is more time consuming. I report the best estimation below.